

Annina Zoé Lieberherr

MSc Interdisciplinary Sciences ETH Zurich



Researcher in theoretical chemistry

Research areas:

Spectroscopy, path integrals, optimal transport

Interests:

Quantum dynamics and photochemistry, accessibility and diversity, project management

Other activities:

Representative for student interests, inclusion, and sustainability

Date of birth: 27.08.1998

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EMPLOYMENT

since
10/2021

DPhil candidate theoretical chemistry

University of Oxford, supervised by David Manolopoulos

Additional activities:

Supervision of two master students during their one-year research project.

Implementation of the sustainability certificate "LEAF Bronze" (2023 and 2024).

Participation in the departmental working group Sustainability.

Representative for graduate students within the chemistry department.

Activity in student organisations:

Treasurer Oxford Women in Chemistry, Event Organiser Oxford University Swiss Society, Women's

Captain Oxford University Amateur Boxing Club, Secretary Oxford University Sustainable Finance Stu-

dent Society, IT and Volunteer Oxford Nightline.

since
10/2022

College lecturer

St Catherine's College, Oxford

Tutorials for 10 students for the lecture "Mathematics for Chemistry", part of the MChem Oxford, in groups of 3 to 4. Organisation and marking of midterms and individual support of students.

05/2022–
08/2022

UNIQ+ event coordinator

University of Oxford

Organisation of an academic and social event schedule for 60 UNIQ+ participants. UNIQ+ gives students from disadvantaged backgrounds the opportunity to do research internships, in preparation for a PhD.

09/2017–
06/2020

Teaching assistant ETH Zurich

Department of Chemistry and Applied Biosciences & Department Mathematics, ETH Zurich

Preparation of weekly exercise classes for groups of 5 to 15 students. Marking of exercise sheets.

DIPLOMAS

2021

M.Sc. Interdisciplinary Sciences (Physics and Physical Chemistry) with distinction

ETH Zurich

Master's thesis at the University of Oxford, supervised by David Tew and Jeremy Richardson.

2020

B.Sc. Interdisciplinary Sciences

ETH Zurich

Bachelor's thesis at the Vrije Universiteit Amsterdam, supervised by Paola Gori-Giorgi and Markus Reiher.

PUBLICATIONS

2024	<i>Vibrational entanglement through the lens of quantum information measures</i> , J. Phys. Chem. Lett. Nina Glaser, Alberto Baiardi, Annina Lieberherr, Markus Reiher
2024	<i>Optimal transport distances to characterise electronic excitations</i> , J. Chem. Theory Comp. Annina Lieberherr, Paola Gori-Giorgi, Klaas Giesbertz
2023	<i>Vibrational strong coupling in liquid water from cavity molecular dynamics</i> , J. Chem. Phys. Annina Lieberherr, Seth Furniss, Joseph Lawrence, David Manolopoulos
2023	<i>Fast quasi-centroid molecular dynamics for water and ice</i> , J. Phys. Chem. B Joseph Lawrence, Annina Lieberherr, Theo Fletcher, David Manolopoulos

AWARDS AND GRANTS

2021–2024	Berrow Foundation Lord Florey Scholarship (Oxford)
2023	Oxford University Innovation Award
2023	Mitacs Research Award (University of Ottawa)
2021	Willi Studer Prize (ETH Zurich)
2020–2021	Hs. H. Günthard Master's Scholarship (ETH Zurich)
2019–2020	University Research Fellowship (VU Amsterdam)

PRESENTATIONS AND POSTERS

May 2024	<i>Vibrational strong coupling in liquid water</i> Virtual International Seminar on Theoretical Advancement (Presentation)
July 2024	<i>Vibrational strong coupling in liquid water from cavity molecular dynamics</i> Seminar for Chemistry and Biomolecular Sciences, University of Ottawa (Presentation)
June 2023	<i>Optimal transport distances for classifying electronic excitations</i> Canadian Mathematical Society Summer Meeting (Presentation)
Dec. 2022	<i>Optimal transport measures for classification of electronic excitations</i> Microsoft AI4Science Lecture (Presentation)
Nov. 2023	<i>Simulating cavity molecular dynamics of liquid water</i> Oxford Women in Chemistry Day (Poster)

LANGUAGES

German	first languages
English	excellent knowledge
French	good knowledge
Mandarin	basic knowledge

PROGRAMMING

Python, Fortran, $\text{\LaTeX}$	excellent knowledge
Mathematica, Julia	good knowledge
C++, HTML	basic knowledge